

ALEX RYSE

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Education

University of Washington

Sep 2026 – Present

Master of Science in Computer Science & Software Engineering

Bothell, WA

Central Washington University

Mar 2026

Bachelor of Science in Computer Science — Minor Mathematics

Ellensburg, WA

- 9 times Dean's List recipient — **GPA: 3.76**

- Relevant Coursework: Data Structures & Algorithms, AI & Machine Learning, Operating Systems, DBMS

Work Experience

AT&T Labs

Jun 2026 – Aug 2026

Software Engineer Intern

Redmond, WA

- Developed **VRPA** (VIAVI Report Processing & Analytics), an end-to-end automation tool utilizing **Flask**, **Streamlit**, and **PostgreSQL** to remotely process, analyze, and report 5G network test data across distributed **Windows Servers**
- Architected an automated **ETL pipeline** and interactive **Plotly** dashboards to parse and visualize massive telemetric log reports containing **50+** CSV files and up to **1GB+** each of raw 5G data, enabling rapid **KPI** root-cause analysis
- Solved a long-standing bottleneck for a **20+** engineer team, cutting overall processing time by over **90%** (1 hr to 5 min)

Nomms

Jun 2025 – Sep 2025

Software Engineer Intern

Ellensburg, WA

- Engineered an **Azure Function** serverless automation solution, reducing user-facing latency by an estimated **98%**
- Participated in stand-up meetings and collaborated closely with a cross-functional team within a full **Microsoft tech stack** including **ASP.NET**, **C#**, and **Bootstrap**, leveraging **Jira** for **Agile development** and issue tracking

Central Washington University

Sep 2025 – Mar 2026

Undergraduate Research Assistant - Cybersecurity

Ellensburg, WA

- Partnered with a professor and **two-student** team to design and implement hands-on **Red/Blue Team lab manuals** for institutional and educational use, leveraging the **MITRE ATT&CK** framework to simulate real-world attacks
- Utilized an **ESXi** lab range to validate attack scenarios using core security tools like **Nmap**, **Snort**, and **Wireshark**

Central Washington University

Apr 2025 – Dec 2025

Teacher Assistant - Advanced Data Structures & Database Management Systems (DBMS)

Ellensburg, WA

- Mentored over **40 students** in **Advanced Data Structures** and **Database Management Systems (DBMS)**, providing actionable feedback on **20+** weekly assignments to **improve code quality** and **enhance understanding**

Projects

CS Study Cat | *GitHub*

.NET Blazor | **SQLite** | **CAG** | **Gemini**

- Spearheaded development as **Project Manager** of a **LLM-powered** educational platform for CWU Computer Science students, leading a **six-person** team through the full **SDLC** and **Agile** sprints that met **100%** of project requirements
- Architected a **Context-Augmented Generation (CAG)** framework to generate grounded **Gemini 3 Flash** responses based on official **CWU CS syllabi**, implementing system-wide **AI guardrails** to maintain strict academic integrity

FirstPass: SWE Job Discovery Engine | *GitHub*

Python | **Flask** | **SQLite** | **Tailwind CSS**

- Developed a multi-process **Python/SQLite** web scraper with a **Flask/Tailwind CSS** UI, for real-time job monitoring
- Leveraged platform automation to streamline application pipelines, directly resulting in **10+** technical SWE interviews

Fitness Tracker | *GitHub*

React | **Flask** | **AWS** | **Docker**

- Designed a full-stack tracker using **React**, **Flask**, and **AWS RDS MySQL** featuring user authentication and tracking
- Optimized application performance by **reducing redundant data fetches** by **80%** with a **React Context API** for global state management and a **JSON Web Token (JWT)** authentication system to secure all backend requests

Competitions

International Drone Competition, Tokyo, Japan | *LinkedIn*

Jul 2025 – Aug 2025

- Collaborated with a **four-person** international team at Tokyo Denki University to develop and apply **Python-based control algorithms** using **OpenCV** for autonomous drone navigation, object detection, and QR code recognition
- Represented Central Washington University as part of a **five-person** delegation that secured **1st** in all competitions

Technical Skills

Languages: Python, Java, C/C++, C#, TypeScript, JavaScript, SQL, PHP

Developer Tools: Docker, Kubernetes, AWS, Azure, PostgreSQL, MySQL, SQLite, Firebase, Git, Linux, Jira

Libraries/Frameworks: Flask, Streamlit, Plotly, React, Node.js, .NET 9, Blazor, ASP.NET Core, Tailwind CSS

AI & Methodologies: ETL Pipelines, CAG, Gemini API, Agile/Scrum, SDLC, MITRE ATT&CK